AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 2 · STUDENT LESSON

Dot products and weighted sums

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Match positions. Multiply. Add. Explain the score.

Calculate signed contributions and preserve their units and interpretation.

Keep one habit: Complete parcel trace → brightness negative connection → cancellation and unit probes → independent three-component calculation.

MAIN LESSON · SECTION 1

How does each input contribute to one score?

> Multiply matching positions before adding;keep score meaning separate from probability.

Learning goals

I can calculate a **dot product** by matching feature positions.

I can explain how a **weight** changes a **score** with other **input**s fixed.

I can distinguish a **weight**ed **score** from a probability or trained model.

Retrieve the fixed input contract

Close your notes first. A rough first answer helps us choose the right starting point; it is not a **score** for your ability.

Without opening the old lesson, explain one key idea from Representing **input**s as vectors. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting explanation

- Under schema [width, height], why must [4, 2] and [2, 4] remain distinct?

> **My starting explanation**

End goal

By the end, I can explain **dot product**s and **weight**ed sums using a traced example and a stated limitation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Calculate a **dot product** by matching feature positions

Explain how a **weight** changes a **score** with other **input**s fixed

Distinguish a **weight**ed **score** from a probability or trained model

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

One matched contribution at a time

> The weights are declared classroom parameters,not fitted results.

Name inputs,weights and the dot product

A small model can combine several measurements into one **score**. Let x = [4, 2] contain fictional width and height measurements in cm. Set **weight**s w = [2, -1].

Multiply matching positions, then add: s = (2 times 4) + (-1 times 2) = 6. The letter x names the **input** vector.

We call this calculation a **dot product**, written w dot x. The vectors must have the same number of components. Each **weight** belongs to one named feature.

These hand-selected **weight**s demonstrate a computation; no **training** has happened. The result 6 is a **score**, not a probability or a claim that this parcel is safe.

Complete signed contributions

Feature	Weight times input	Contribution
Width	2 times 4	8
Height	-1 times 2	-2
Total	8 plus -2	6

Keep units compatible

The coefficients carry **score**-per-cm units in this example, making both **contribution**s compatible **score** units. A **weight**ed sum needs a consistent definition of units.

Identify supplied and missing facts

List the given facts and one fact that is not supplied.

MAIN LESSON · SECTION 3

Trace every signed product before combining it

Zero can mean cancellation between nonzero contributions.

Complete four-input trace

Parcel input [width_cm, height_cm]	Width contribution 2×width	Height contribution−1×height	Total score
A[4, 2]	8	-2	6
B[2, 4]	4	-4	0
C[3, 3]	6	-3	3
Same-width changed height[4, 3]	8	-3	5

Keep w = [2, -1] fixed. For A = [4, 2], the **score** is 6. For B = [2, 4], it is 4 - 4 = 0. For C = [3, 3], it is 6 - 3 = 3. Swapping **input** positions without swapping their matching **weight**s changes the computation.

A zero **score** can come from cancellation, even when neither **input** is zero.

Runnable matched-length dot product

```
def dot(weights, inputs):
    if len(weights) != len(inputs):
        raise ValueError("Mismatched dimensions")
    return sum(w * x for w, x in zip(weights, inputs))

for row in [[4, 2], [2, 4], [3, 3]]:
    print(dot([2, -1], row))  # 6, 0, 3
```

The length check matters: Python zip alone stops at the shorter list and could silently omit a feature. The function assumes numeric **input**s already validated under the schema.

First predict the three printed values on paper, then compare.

Connect each visual arrow to a product

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Change one input while keeping its coefficient fixed

> Predict the sign and size of the score change.

Current cancellation case and response

- For w = [2, -1] and x = [2, 4], explain why **score** zero does not mean the **input** contains no information.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured brightness change

- Change: Increase only the right **input** from 5 to6.

▼ **Check after predicting**

Prediction: The **score** becomes 6-6=0.

Reason: The right **weight** is-1, so adding 1 to that **input** lowers the **score** by1.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

See addition and subtraction as separate contributions

> The negative connection lowers the total as its input increases.

Complete brightness facts and separate weights

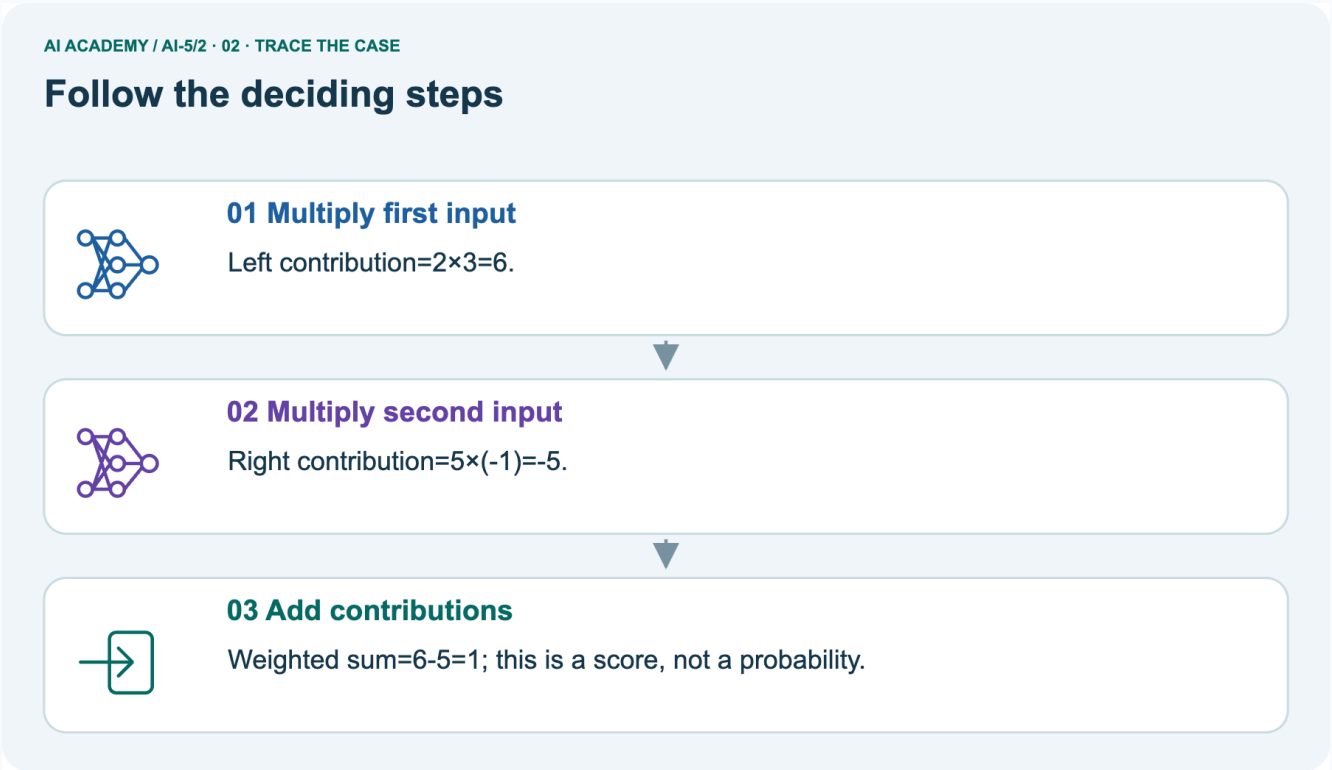
Draw each **input**'s **contribution** to a toy **weight**ed brightness **score**.

Inputs [2, 5]; **weight**s [3,-1]; no bias. **Score** is 2×3+5×(-1).

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

This brightness picture uses x=[2, 5], w=[3, −1], no bias:contributions 6 and−5, total 1. Increasing only the right input to6 changes that contribution to−6, total 0. The input still has nonzero measurements; the contributions cancel. Main parcel examples instead use w=[2, −1], so do not swap the two weight lists. A score can be negative, zero or positive; no probability interpretation or training is supplied.

Trace both signed contributions



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Multiply first input	Left contribution=2×3=6.
2. Multiply second input	Right contribution=5×(-1)=-5.
3. Add contributions	Weighted sum=6-5=1; this is a score, not a probability.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

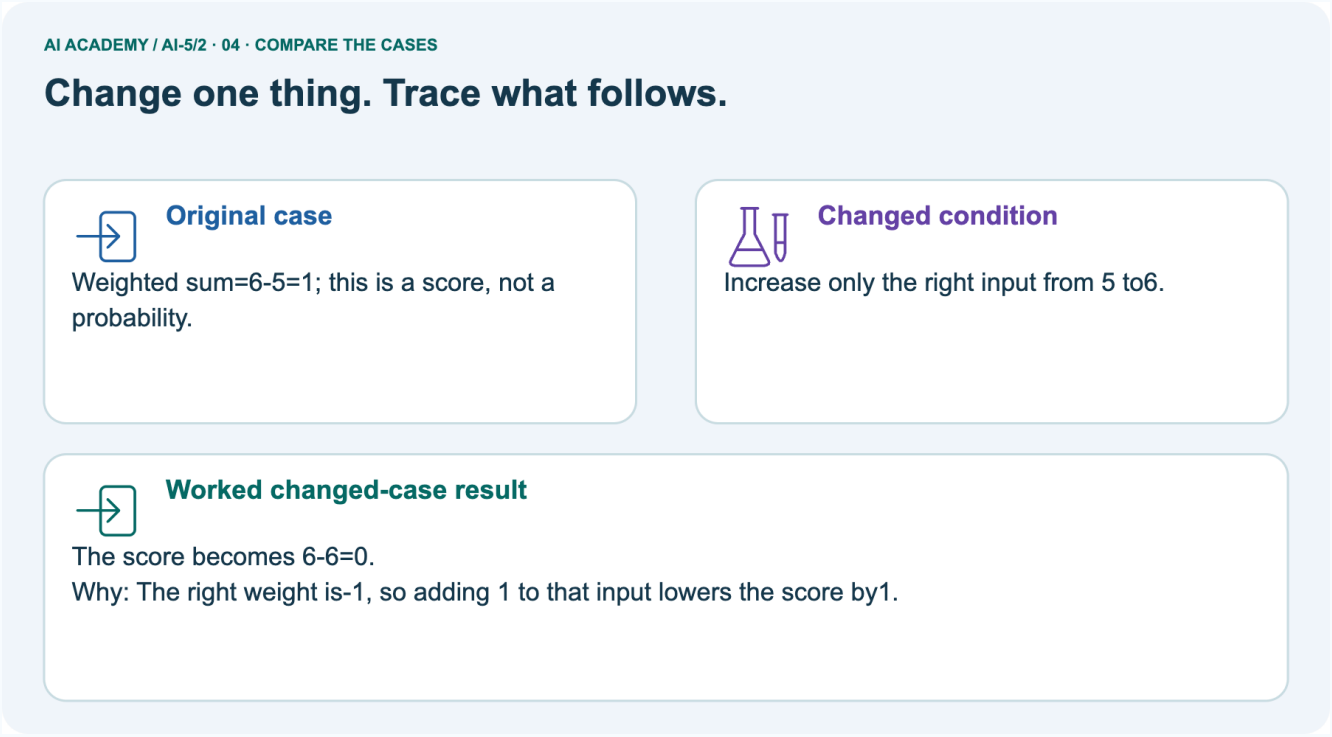
A **weight** controls how strongly an **input** contributes to the **score**. The left connection adds three times its measurement; the negative right connection subtracts its measurement. Calculate each **contribution** before adding them.

This **dot product** is a signed **score**; it becomes a probability only if a separate, justified transformation and interpretation are supplied.

Predict the increased right input

- Change one thing: Increase only the right **input** from 5 to6. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain score and training limits

- These declared **weight**s are hand-chosen toy parameters; no network **training** or calibrated probability is claimed.

- Read the labelled picture: Which connection explains why more right brightness lowers this **score**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Try the main parcel height change

Multiply matching components then add

POSITION	CONTRIBUTION RULE	WORKED CONTRIBUTION
1 width	2 times width	2 times 4 gives 8
2 height	-1 times height	-1 times 2 gives -2
Add both	8 plus -2	Score 6

The first two rows are separate contributions. The last row combines them; multiplying the final input sum by the final weight sum would be a different operation.

Keep w = [2, -1]. Change [4, 2] to [4, 3]. Compute both contributions and explain the score change.

▼ **Earlier wording, alternatives and picture transcripts**

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Dot products and weighted sums

The facts we will use

AI ACADEMY / AI-5/2

Weighted brightness score

Input 2 x weight 3 = 6

Input 5 x weight -1 = -5

Add the products and bias 0: total = 1

Keep the total (identity) -> output 1

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/2

Dot products and weighted sums

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Multiply first input

Left contribution=2 x 3=6.



2. Multiply second input

Right contribution=5 x (-1)=-5.



3. Add contributions

Weighted sum=6-5=1; this is a score, not a probability.

CHANGE ONE THING

Increase only the right input from 5 to6.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Repair unit errors and overclaims about weight signs

> Name the controlled feature and its unit before interpreting a coefficient.

Fixed-input change and unit-preserving coefficients

Holding other **input**s fixed, raising width by 1 cm raises this **score** by 2; raising height by 1 cm lowers it by 1. A negative **weight** reverses the direction of that feature **contribution**.

It does not label the feature as bad, and its sign alone does not explain cause in the world.

Unit changes affect coefficients. If height is written in mm instead of cm while width remains in cm, use height **weight** -0.1 instead of -1 to preserve the same **score**s.

For [4 cm, 20 mm], 2 times 4 plus -0.1 times 20 still gives 6. Raw coefficient magnitudes cannot establish feature importance when units differ.

Try the cm/mm repair

- Width stays in cm and height changes to mm. For the worked parcel [4 cm, 20 mm], give the revised **weight**s and **score**.

> **State one repair and the checks you would rerun.**

▼ **Earlier wording, alternatives and picture transcripts**

Locate the earliest unsupported step

Calculate and explain with less help

> A matching total without matching products is incomplete evidence.

Try a new score and height change

- For w = [2, -1] and x = [3, 5], write both **contribution**s and their sum.

> **Keep w = [2, -1]. Change [4, 2] to [4, 3]. Compute both contributions and explain the score change.**

> **Explain your prediction to a partner before comparing results.**

Supported W2record

Use workbook W2 for the connected example "**Dot product**s and **weight**ed sums". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge multiply-the-sums reasoning

- Use the worked case for **Dot product**s and **weight**ed sums. Proposed explanation: “Multiply the sums instead”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the pictured negative connection

- Which connection explains why more right brightness lowers this **score**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

Reserve the independent three-component case

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Explain cancellation

Check your reasoning

Explain the picture

Investigate cancellation,dimension guards and unit-preserving change

> The complete probes distinguish valid arithmetic from a valid task input.

Complete current task and runnable function

For w = [2, -1] and x = [2, 4], explain why **score** zero does not mean the **input** contains no information.

```
def dot(weights, inputs):
    if len(weights) != len(inputs):
        raise ValueError("Mismatched dimensions")
    return sum(w * x for w, x in zip(weights, inputs))

for row in [[4, 2], [2, 4], [3, 3]]:
    print(dot([2, -1], row))  # 6, 0, 3
```

All supplied probes

Complete dot-product probe	Expected output / reason
Normal w[2, –1], x[3, 5]	6–5=1
Cancellation w[2, –1], x[2, 4]	4–4=0; input still nonzero
Zero coefficient w[2, 0], x[3, 5]	6+0=6; second feature has no effect on this score
Mismatched w[2], x[3, 5]	ValueError:Mismatched dimensions
Missing value w[2, –1], x[3, None]	Outside numeric-input contract; validate/request before calling, not substitute 0
Height converted to mm	w[2, –0.1], x[4, 20]→6, matching original cm score

Complete dot-product probe	Expected output / reason
Both lists empty	Python returns 0; this numerical edge result does not represent this week's two-feature parcel schema

Predict,run or trace,and interpret

1. Use the complete dot function from section 3. Predict each supplied probe, then run or trace it and retain products, output/error and interpretation. Treat missing measurements as an earlier schema problem. The function checks equal length only; it assumes validated numeric values and permits an empty sum. For the declared parcel task, require the two named components before calling.

   Compare[4, 2]and[5, 4]:both score 6, so a later step receiving only this score cannot tell them apart.

2. On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Keep the evidence ledger

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or "not run"	
First mismatch and one explanation	
Boundary or missing input and expected response	

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep "not run" if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: a new three-component dot product

> Match every input with its own coefficient and explain the single changed component.

Complete selected inputs and tasks

Use x = [3, 1, 2] and w = [-2, 4, 0.5]. Give each **contribution** and the total. Increase only the second component from 1 to 2 and compute the new total. Explain why a two-**weight** list is invalid for this **input**.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Compute a new dot product independently

Attempt this without hints

Use x = [3, 1, 2] and w = [-2, 4, 0.5]. Give each **contribution** and the total. Increase only the second component from 1 to 2 and compute the new total. Explain why a two-**weight** list is invalid for this **input**.

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Add signed contributions to the project record

> Keep the input schema,declared weights and score interpretation together.

Project artifact and table

Add a labelled **dot product**s and **weight**ed sums **evidence** sheet to Image-pattern Network Investigation. Show how you can calculate a **dot product** by matching feature positions.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Keep the actual evidence and limits

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish with one signed contribution and one unsupported claim

> Retrieve the procedure before opening the worked trace.

Explain without notes

- Close your notes. Explain this goal in your own words: Calculate a **dot product** by matching feature positions

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use the mathematical words precisely**

REFERENCE · SECTION 5

Reference · Use the mathematical words precisely

> Connect each term to a value or step in the current example.

Vocabulary and your explanation

Word	Meaning
Weight	A numerical coefficient multiplying one input component
Dot product	The sum of products of corresponding vector components
Contribution	The product of one input component and its matching weight
Score	A numerical output whose interpretation must be specified

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- For w = [2, -1] and x = [3, 5], write both **contribution**s and their sum.
- Width stays in cm and height changes to mm. For the worked parcel [4 cm, 20 mm], give the revised **weight**s and **score**.
- Verify that [4, 2] and [5, 4] both **score** 6 with w = [2, -1]. Can a later step receiving only this **score** distinguish those two records?

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the representation and its limits

REFERENCE · SECTION 12

Reference · Check the representation and its limits

> Use after practice;reserve the new three-component dot product and changed second input for the first individual attempt.

Current explanation and response

- A **weight**ed sum can combine **evidence**, but it discards information: [4, 2] and [5, 4] both produce **score** 6 under these **weight**s. A downstream rule receiving only that **score** cannot recover which of these vectors was used.

  To classify, we would also need a stated decision rule and labelled evaluation examples.

- Next week adds an offset called a bias and states exactly how a **score** becomes a class **prediction**.

- Verify that [4, 2] and [5, 4] both **score** 6 with w = [2, -1]. Can a later step receiving only this **score** distinguish those two records?

A teammate calls score 6 a 60 percent chance of class 1. What is missing from that claim?

My limited claim and a useful next check

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Multiply matching components then add

POSITION	CONTRIBUTION RULE	WORKED CONTRIBUTION
1 width	2 times width	2 times 4 gives 8
2 height	-1 times height	-1 times 2 gives -2
Add both	8 plus -2	Score 6

The first two rows are separate contributions. The last row combines them; multiplying the final input sum by the final weight sum would be a different operation.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Draw each **input**'s **contribution** to a toy **weight**ed brightness **score**.

By the end, I can	How I will show it
Calculate a dot product by matching feature positions	Explain the deciding step, show a trace or test, and state one limit.
Explain how a weight changes a score with other inputs fixed	Explain the deciding step, show a trace or test, and state one limit.
Distinguish a weighted score from a probability or trained model	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Draw each **input**'s **contribution** to a toy **weight**ed brightness **score**.
- Which labelled step matches this detail? **Weight**ed sum=6-5=1; this is a **score**, not a probability.
- 1. Multiply second **input**
- 2. Add **contribution**s
- 3. Multiply first **input**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Add **contribution**s **Weight**ed sum=6-5=1; this is a **score**, not a probability. A **weight** controls how strongly an **input** contributes to the **score**.

The left connection adds three times its measurement; the negative right connection subtracts its measurement. Calculate each **contribution** before adding them.

This **dot product** is a signed **score**; it becomes a probability only if a separate, justified transformation and interpretation are supplied.

Changed-case question and optional hint

- Changed-case practice: Increase only the right **input** from 5 to6. Which conclusion is supported by the supplied facts?
- 1. The **score** becomes 6-6=0.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The **score** becomes 6-6=0. The right **weight** is-1, so adding 1 to that **input** lowers the **score** by1.

Repair and retain the record

Review Representing **input**s as vectors (ai-5/1). Goal: Encode a record using a fixed feature order and units. Short practice: Why does equal total brightness not make A and B identical **input**s?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ Reference · Six-picture concept journey

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Dot products and weighted sums: the case

AI ACADEMY / AI-5/2 · 01 · LOOK AT THE FACTS

Dot products and weighted sums: the case

The situation

Draw each input's contribution to a toy weighted brightness score.

Supplied evidence

Inputs [2, 5]; weights [3,-1]; no bias. Score is 2×3+5×(-1).
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain dot products and weighted sums using a traced example and a stated limitation.

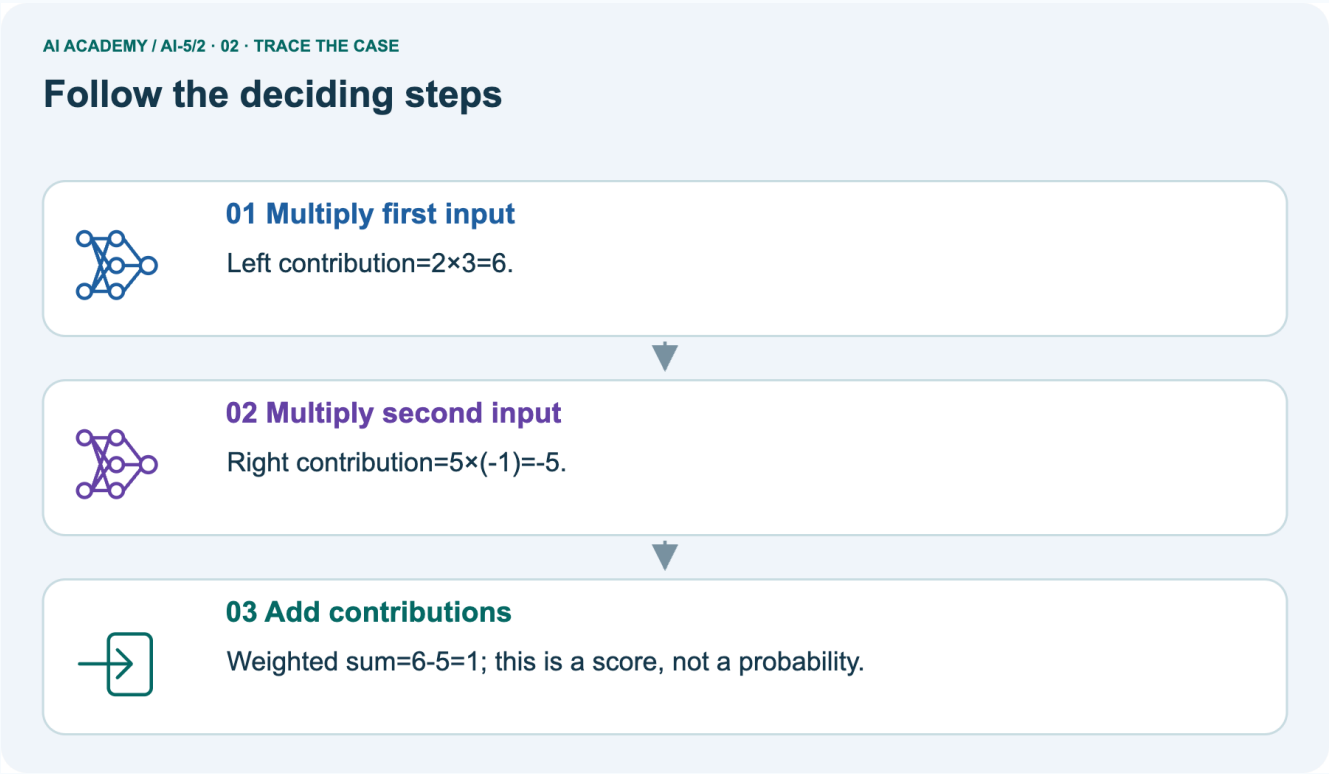
Follow the labelled picture with the matching explanation. Open the vector diagram

Draw each **input**'s **contribution** to a toy **weight**ed brightness **score**. Supplied facts: **Input**s [2, 5]; **weight**s [3,-1]; no bias. **Score** is 2×3+5×(-1).

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

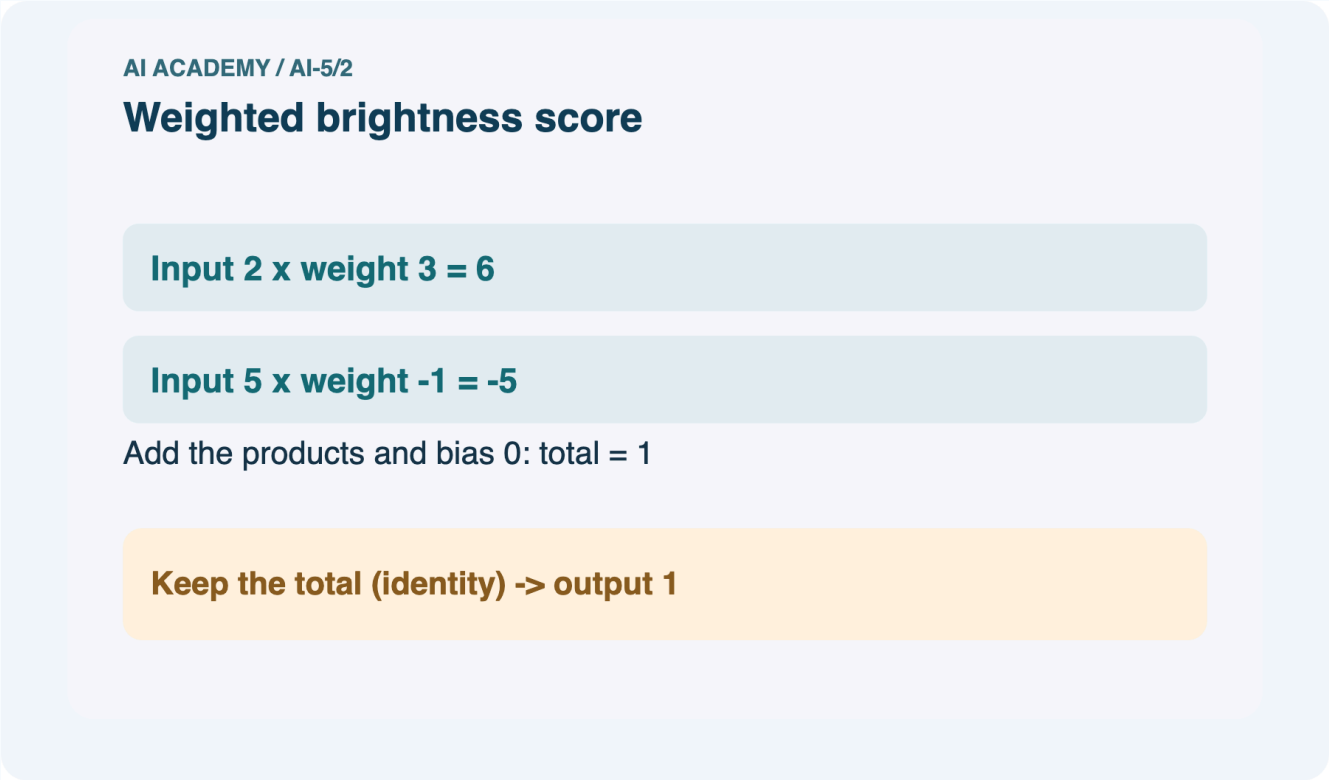


Follow the labelled picture with the matching explanation. Open the vector diagram

Multiply first **input**: Left **contribution**=2×3=6. → Multiply second **input**: Right **contribution**=5×(-1)=-5. → Add **contribution**s: **Weight**ed sum=6-5=1; this is a **score**, not a probability.

Point to each numbered stage and explain which recorded **input** it uses.

03. Weighted brightness score

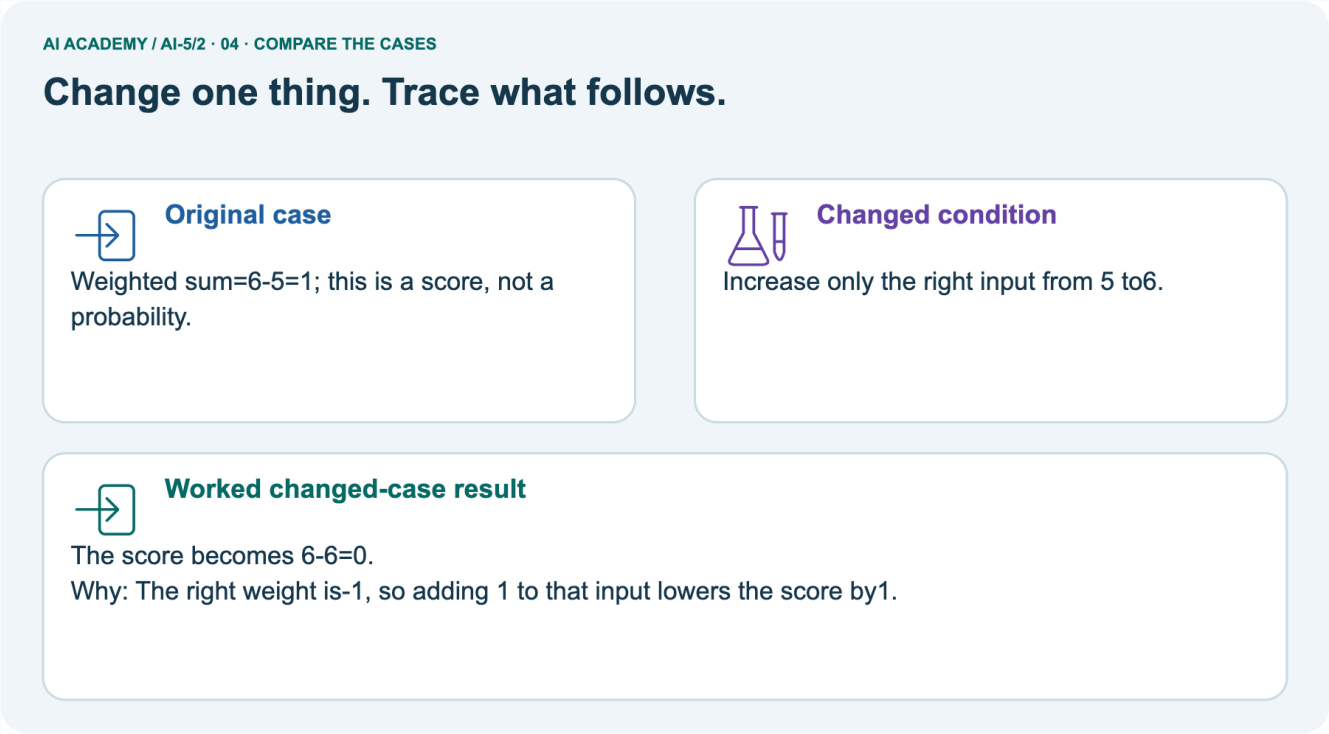


Follow the labelled picture with the matching explanation. Open the vector diagram

Weighted brightness **score**. **Input**s: [2, 5]; **weight**s: [3, -1]; bias: 0; activation: identity; **output**: 1.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



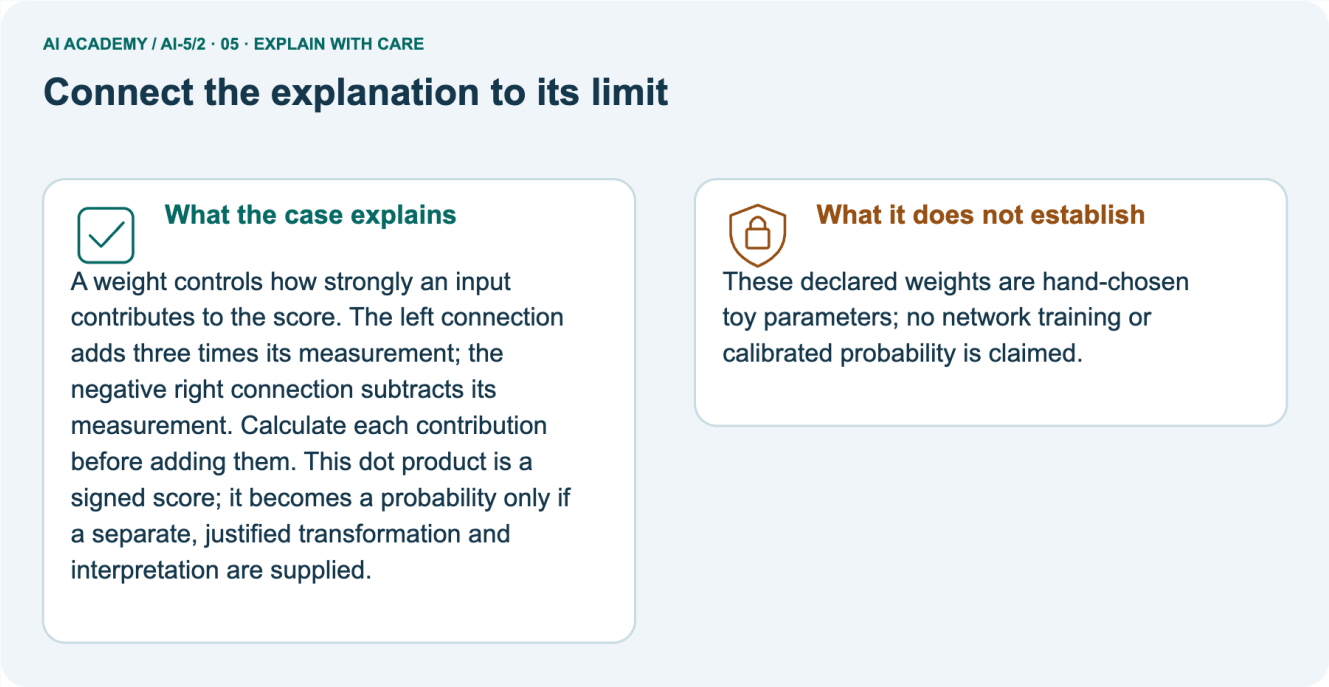
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: **Weight**ed sum=6-5=1; this is a **score**, not a probability. Changed condition: Increase only the right **input** from 5 to6. Expected result: The **score** becomes 6-6=0.

Reason: The right **weight** is-1, so adding 1 to that **input** lowers the **score** by1.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

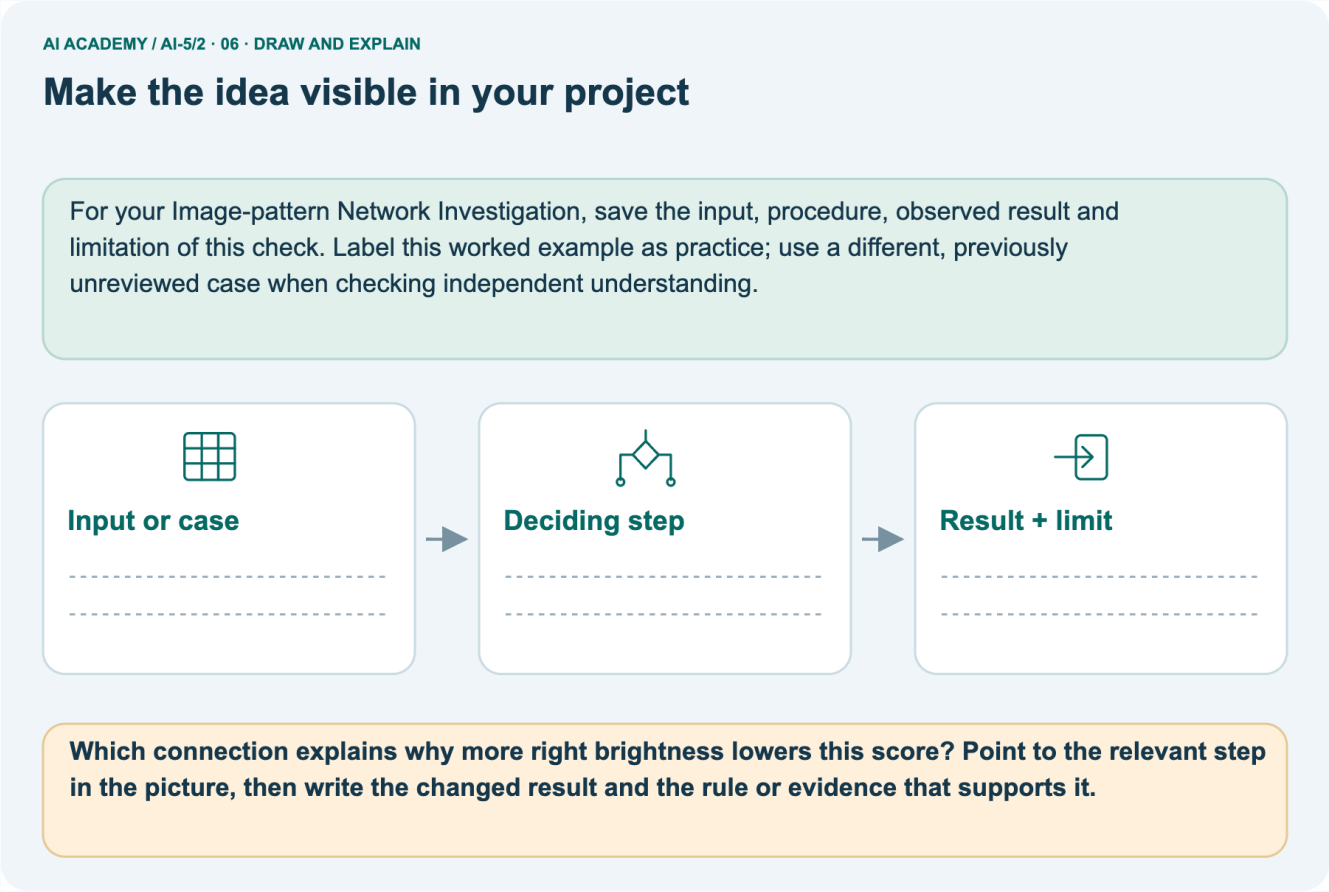
A **weight** controls how strongly an **input** contributes to the **score**. The left connection adds three times its measurement; the negative right connection subtracts its measurement. Calculate each **contribution** before adding them.

This **dot product** is a signed **score**; it becomes a probability only if a separate, justified transformation and interpretation are supplied.

Limit: These declared **weight**s are hand-chosen toy parameters; no network **training** or calibrated probability is claimed.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which connection explains why more right brightness lowers this **score**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Dot products and weighted sums: the case

02. Follow the deciding steps

03. Weighted brightness score

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.